

From Rate Suppression to Outcome Starvation: A Finite-Width Causal Theory of Recurrent Learning

Anonymous authors

ICLR 2027 submission draft — 24 August 2026

Abstract

A strong predictive cue can accelerate optimization while preventing a weaker but independently predictive cue from being learned. We study this gradient-starvation question in recurrent binary classifiers using a paired causal intervention: a both-feature trajectory is compared with a same-initialization trajectory in which only the strong input channel is ablated. Our contribution is a finite-width package rather than a claimed general dynamical mean-field theory (DMFT). First, we derive the exact cross-entropy response flow through a response-logit cross-kernel and identify the additional assumption under which it reduces to a closed two-mode equation $\dot{M} = Gg$. Second, after a transfer-to-suppression rate crossing, we prove that outcome starvation occurs if and only if the negative post-crossing drift area exhausts the earlier transfer advantage. Third, for a noiseless rank-one cue law, an ordering-invariant log drift ratio gives sufficient conditions for a unique rate crossing. We also prove transverse hitting-time stability and an exact six-scalar, zero-disorder/zero-lag wide limit with an explicit $9/(N\epsilon^2)$ initialization bound. Finally, we derive Counterfactual Drift Correction (CDC), the unique uncapped minimum-norm correction that matches a weak-only drift target while preserving instantaneous strong-response drift, and prove a local $O(\eta^2)$ one-step deviation bound.

Controlled dense-linear experiments exhibit positive weak-learning AUC gaps in 32/32 seeds at lags 0, 2, and 4. Under the final direct-autograd nonlinear protocol, however, tanh transfer-to-starvation is seed-dependent (3/8 causal certificates), while GRU has 5/8 response crossings but 0/8 causal certificates because the weak-only model never passes the learnability gate. CDC uniquely preserves instantaneous strong drift, but does not improve weak rescue or final strong retention over all shadow-based ablations. These corrected results delimit the claim: the causal and finite-width theorems are established; a general positive-disorder/positive-lag optimization-time DMFT, an analytic phase boundary, and universal nonlinear crossover remain open.

1 Introduction

Cross-entropy training can favor a subset of predictive features strongly enough that other useful features are learned slowly or not at all. This phenomenon was formalized as gradient starvation by Pezeshki et al. [2021]. Recurrent networks add a temporal geometry: cue strength, lag, shared recurrent parameters, and evolving feature response geometry all affect which signal receives gradient. The resulting mechanism is easy to overstate. A weak-feature drift can decline without changing sign; its relative drift can change sign without the weak response falling behind; and a response gap is not evidence of causal starvation if the weak-only counterfactual is itself unable to learn.

We address these distinctions with a paired intervention. The both-feature and weak-only models share initialization, labels, samples, weak coordinates, and noise; only the strong input channel is set to zero in the weak-only condition. We track common feature-response functionals in both trajectories. This setup separates three statements:

1. *relative transfer*: the strong cue initially increases the weak response rate;
2. *rate suppression*: the both-feature weak response later grows more slowly than its counterfactual;
3. *causal outcome starvation*: the both-feature weak response falls behind and the weak-only counterfactual demonstrably learns.

The distinction is both mathematical and empirical: the final GRU experiment has response crossings but fails the causal learnability gate in every seed.

Contributions.

1. We prove an exact finite-width response-flow identity for logistic gradient flow. A closed projected equation $\dot{M} = Gg$ follows only under exact, parameter-independent logit realization; nonlinear probe responses carry an explicit residual.
2. We give a necessary-and-sufficient negative-tail-area criterion converting a unique rate crossover into an outcome crossing, together with quantitative crossing-time bounds.
3. For noiseless rank-one cues, we derive an ordering-invariant log drift ratio and sufficient conditions for a unique rate crossover. Unlike a non-canonical additive geometry/field split, the ratio has a fixed meaning.
4. We prove stability of transverse first hitting times and an exact zero-disorder, zero-lag six-scalar closure with finite-horizon convergence in probability and explicit constants.
5. We prove three local CDC results: exact instantaneous strong-drift preservation, unique uncapped minimum-norm optimality when feasible, and a local $O(\eta^2)$ one-step strong-response deviation bound.
6. We provide executable theorem contracts and a claim-audited experiment suite. The corrected nonlinear evidence is reported as seed-dependent and the unsuccessful exploratory search is retained as a negative result.

Claim boundary. We do not claim that recurrent networks universally undergo transfer-to-starvation, that a general joint CE-RNN optimization-time DMFT has been solved, or that a crossover time or analytic ρ_c is independently predicted. The general DMFT solver remains deliberately unimplemented where its proof obligations are unmet.

2 Related work

Gradient starvation. Pezeshki et al. [2021] identify feature starvation under cross-entropy and motivate Spectral Decoupling. Our focus is different: recurrent temporal features, a paired strong-channel intervention, exact finite-width response dynamics, and the logical gap between rate suppression and outcome starvation.

Learning dynamics in recurrent networks. Ger and Barak [2026] derive low-dimensional overlap learning dynamics for low-rank RNNs and show that overlaps invisible to the loss can retain training history. This directly cautions against closing a recurrent learning theory using only visible response coordinates. Clark et al. [2026] develop a mean-field account of structure and disorder in task-trained recurrent circuits in a long-time Langevin/Gibbs setting. That equilibrium construction is distinct from the paired, noiseless optimization-time gradient flow studied here. Accordingly, we claim neither the first learning theory for RNNs nor a completed general RNN DMFT; our novelty lies in the causal pairing and finite-width rate-to-outcome results.¹

Gradient surgery. PCGrad projects conflicting task gradients [Yu et al., 2020]; Bloop stabilizes an auxiliary projection using an exponential moving average [Hsieh et al., 2024]. Projection itself is not our novelty. CDC protects the gradient of a theory-defined strong-feature response and takes its target from a matched causal weak-only shadow. This stronger information setting must be reported: CDC requires a constructible ablation, a second model, lockstep training, and full-batch gradients.

¹The descriptions of the 2026 sources were paraphrased from the linked public sources. Content was rephrased for compliance with licensing restrictions.

3 Causal setup and definitions

3.1 Paired logistic gradient flow

For condition $C \in \{B, W\}$, let $\theta_C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L_C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (1)$$

The two conditions share initialization and all data except that the strong channel is zeroed under W. Let M_s^C, M_w^C be common differentiable strong and weak response functionals. Shared initialization gives

$$\Delta(0) = 0, \quad \Delta(\tau) = M_w^B(\tau) - M_w^W(\tau). \quad (2)$$

Optimization time τ is distinct from sequence index k .

The controlled dense linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (3)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$, $B_{ia}(0) \sim \mathcal{N}(0, 1/N)$, and $c_i(0) \sim \mathcal{N}(0, 1/N)$. The width scaling of B is essential for a non-divergent projected metric. We additionally study tanh RNNs and GRUs. For nonlinear models the two responses are common odd symmetric unit-probe contrasts, $[f(x_a^+) - f(x_a^-)]/2$, not assumed additive coordinates of every data logit.

3.2 Causal labels and metrics

Let

$$d(\tau) = \Delta'(\tau) = F_w^B(\tau) - F_w^W(\tau). \quad (4)$$

We call $d > 0$ relative transfer and $d < 0$ relative rate suppression. An outcome crossing occurs when Δ reaches zero after a positive excursion and then becomes negative. We reserve *causal starvation* for outcome suppression when the weak-only trajectory also reaches a preregistered target β within horizon H . If the weak-only model does not reach β , the result is indeterminate rather than starvation.

We report first-hitting-time delay when both hits are observed. The horizon-wide weak AUC gap is

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) \, d\tau, \quad (5)$$

computed by the logged-grid trapezoid rule. GSI-5 is retained as a margin-weight diagnostic, not as the definition of starvation.

4 Exact finite-width response dynamics

Theorem 1 (Universal response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (6)$$

No width limit or two-mode realization is required.

Corollary 2 (Closed projected flow under exact realization). *If the signed logits admit the exact parameter-independent representation $r_i^C = \sum_b z_{ib}^C M_b^C$, then*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n_C} \sum_i z_{ib}^C \sigma(-z_i^C \cdot M^C). \quad (7)$$

If $r_i^C = z_i^C \cdot M^C + r_i^\perp$, the exact equation instead contains

$$R_a^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle, \quad \dot{M}_a^C = (G^C g^C)_a + R_a^C. \quad (8)$$

For channel-localized modes of the dense linear RNN, differentiation across the disjoint (c, B, W) blocks gives the exact finite-width geometry

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (9)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (10)$$

The proofs are in Appendix A. Equation (7) is exact for the zero-background-noise linear synthetic model. For tanh/GRU probes, we use Theorem 1's direct autograd drift as primary and report R ; projected and matched-state decompositions are diagnostics only.

5 From rate suppression to outcome starvation

Theorem 3 (Necessary and sufficient tail-area criterion). *Let Δ be absolutely continuous on $[0, H]$, with $\Delta(0) = 0$. Suppose there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Define*

$$A_+ = \int_0^{\tau_d} d(u) du = \Delta(\tau_d) > 0, \quad N(t) = \int_{\tau_d}^t -d(u) du. \quad (11)$$

Then $\Delta(t) = A_+ - N(t)$ for $t \geq \tau_d$. Hence equality is reached by H if and only if $N(H) \geq A_+$, and strict outcome starvation occurs by H if and only if $N(H) > A_+$. If $d < 0$ on each nontrivial post-crossing interval, the later equality is unique whenever it exists.

This result blocks a common but false inference: a $+$ $\rightarrow$ $-$ drift crossing locates a local maximum of Δ ; it does not force Δ to return to zero. The post-crossing negative area must repay the positive area already accumulated. Causal starvation further requires weak-only learnability.

Corollary 4 (Response-crossing bounds). *Under Theorem 3, if $d(t) \leq -\kappa < 0$ on $[t_0, H]$ with $t_0 > \tau_d$, then, when the right side lies within the horizon,*

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N(t_0)]_+}{\kappa} \leq t_0 + \frac{A_+}{\kappa}. \quad (12)$$

If $d(\tau_d) = 0$ and $d'(t) \leq -\lambda < 0$ on $[\tau_d, \tau_d + r]$, then $N(t) \geq \lambda(t - \tau_d)^2/2$; when $A_+ \leq \lambda r^2/2$,

$$\tau_{\text{response}} \leq \tau_d + \sqrt{2A_+/\lambda}. \quad (13)$$

5.1 An ordering-invariant rank-one criterion

Assume deterministic positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, $\rho > 0$, and exact two-mode realization. Define

$$u_B = \rho M_s^B + M_w^B, \quad s_B = \sigma(-u_B), \quad (14)$$

$$u_W = M_w^W, \quad s_W = \sigma(-u_W), \quad (15)$$

$$H_B = \rho G_{ws}^B + G_{ww}^B, \quad H_W = G_{ww}^W, \quad (16)$$

$$Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0. \quad (17)$$

Theorem 5 (Log drift-ratio identity). *The weak drifts factor as $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$. Where $H_B, H_W > 0$, define*

$$\Psi = \log \frac{F_w^B}{F_w^W} = \log \frac{s_B}{s_W} + \log \frac{H_B}{H_W}. \quad (18)$$

Then $\text{sign}(d) = \text{sign}(\Psi)$ and

$$\Psi' = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \left(\log \frac{H_B}{H_W} \right)'. \quad (19)$$

Corollary 6 (Unique rate crossing). *Suppose on $[t_0, H]$ that $H_B, H_W > 0$, $\Psi(t_0) > 0$, $\Psi'(t) \leq -\kappa < 0$, and $H - t_0 \geq \Psi(t_0)/\kappa$. Then exactly one rate crossing occurs in $(t_0, H]$ and*

$$\tau_d \leq t_0 + \Psi(t_0)/\kappa. \quad (20)$$

A sufficient primitive condition is

$$s_B(1 - s_B)Q_B \geq a, \quad s_W(1 - s_W)H_W \leq b, \quad (\log(H_B/H_W))' \leq c, \quad a - b - c \geq \kappa > 0. \quad (21)$$

The bounds must be established independently to be predictive. Evaluating Ψ on a completed trajectory is certification, not prediction. Noisy cues generally break the rank-one factorization, and even a rate crossing still requires Theorem 3 for an outcome crossing.

5.2 Hitting-time stability

Corollary 7 (Transverse first-hit stability). *Let continuous x_N, x on $[0, H]$ satisfy $\|x_N - x\|_\infty \leq \epsilon_N$. Let $\tau \in (0, H)$ be the first upward crossing of β , separated from earlier times, and suppose $x'(t) \geq \kappa > 0$ on $[\tau - r, \tau + r]$. For sufficiently small ϵ_N ,*

$$|\tau_N - \tau| \leq \epsilon_N/\kappa. \quad (22)$$

A logged grid of spacing h_N adds h_N . For paired delays,

$$|\Delta\tau_N - \Delta\tau| \leq \epsilon_{B,N}/\kappa_B + \epsilon_{W,N}/\kappa_W + 2h_N. \quad (23)$$

This transfers an independently proved uniform trajectory limit; it does not prove that limit.

6 A solvable zero-disorder wide limit

Set recurrent gain $g = 0$, lag separation zero, and background noise zero in the dense linear model. Write the input columns as b_s, b_w . The recurrent block is inert, $M_a = c \cdot b_a$. Let $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$.

Theorem 8 (Every-width six-scalar closure). *At every width,*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad (24)$$

$$\dot{w} = 2 \sum_b g_b M_b, \quad G_{ab} = u_{ab} + \delta_{ab} w. \quad (25)$$

Theorem 9 (Finite-horizon deterministic wide limit). *Let all coordinates of b_s, b_w, c be mutually independent $\mathcal{N}(0, 1/N)$ and*

$$q_N = (M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w), \quad q_0 = (0, 0, 1, 0, 1, 1). \quad (26)$$

For every $\epsilon > 0$,

$$\mathbb{P}(\|q_N(0) - q_0\|_\infty > \epsilon) \leq \min \left\{ 1, \frac{9}{N\epsilon^2} \right\}. \quad (27)$$

On every fixed horizon $[0, H]$, let L_H be a vector-field Lipschitz constant on a common compact tube containing the finite- and infinite-width trajectories. Then

$$\sup_{\tau \leq H} \|q_N(\tau) - q(\tau)\|_\infty \leq e^{L_H H} \|q_N(0) - q_0\|_\infty, \quad (28)$$

and

$$\mathbb{P} \left(\sup_{\tau \leq H} \|q_N(\tau) - q(\tau)\|_\infty > \delta \right) \leq \min \left\{ 1, \frac{9e^{2L_H H}}{N\delta^2} \right\}. \quad (29)$$

Thus $q_N \rightarrow q$ uniformly on fixed horizons in probability, with initialization scale $O_{\mathbb{P}}(N^{-1/2})$.

This theorem is intentionally singular: it is a genuine large-width result only at zero disorder and zero lag. It is not a positive-disorder DMFT.

Table 1: Executable contracts in `src/gradient_starvation/theory.py`.

Contract	Mathematical object	Scope guard
<code>projected_statistics</code>	direct Eq. (6), Gg , residual	direct drift primary for nonlinear probes
<code>equal_time_drift_difference</code>	Δ' and two projected product orderings	exposes non-canonical attribution
<code>rank_one_drift_ratio</code>	Theorem 5 factors and Ψ	caller supplies noiseless/exact hypotheses
<code>discrete_crossover_certificate</code>	telescoping finite-step tail areas	learnability gate required for causal label
<code>transverse_hitting_time_error_bound</code>	$\epsilon/\kappa + h$	caller supplies uniform error/transversality
<code>zero_disorder.*failure_bound</code>	Eqs. (27), (29)	caller supplies valid common-tube L_H
<code>cdc_finite_step_deviation_bound</code>	Eq. (32)	caller supplies local smoothness

7 Counterfactual Drift Correction

At a both-feature state θ , let

$$a = \nabla m_s(\theta), \quad p = \nabla m_w(\theta), \quad v = -\nabla L(\theta), \quad F_s = a \cdot v, \quad F_w = p \cdot v. \quad (30)$$

A same-initialization weak-only shadow supplies F_w^W . Define

$$q = \begin{cases} p - \frac{p \cdot a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \delta = [F_w^W - F_w]_+, \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (31)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible. The theorems assume full-batch gradients, plain SGD, zero weight decay, and no clipping. A coefficient cap preserves orthogonality but can lose target attainment and uncapped optimality.

Theorem 10 (CDC local guarantees). *The construction in Eq. (31) has the following properties.*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly, including when α is capped.
2. **Minimum norm:** for the feasible uncapped problem, $d^* = \alpha q$ is the unique minimum-norm correction satisfying $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$.
3. **Local finite-step bound:** if ∇m_s is L -Lipschitz on a convex neighborhood containing both step segments, then

$$|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2). \quad (32)$$

If $\|v\| \leq V$, $\delta \leq D$, and $\|q\| \geq q_{\min} > 0$, the right side is at most

$$\frac{L}{2} \eta^2 [V^2 + (V + D/q_{\min})^2]. \quad (33)$$

The result is one-step and local. It is not a trajectory guarantee and does not imply final strong-response retention.

8 Executable theorem contracts

The implementation makes theorem scope observable rather than silently extending it. Table 1 lists the primary contracts. Unit tests check these software identities and edge cases; tests are not mathematical proofs.

Every general positive-disorder/positive-lag solver path that would require an unproved closure raises `NotImplementedError`. This is a scientific guard, not missing numerical polish.

Table 2: Final synthetic experiment configurations. “Shadow” methods train a second weak-only model and therefore use roughly twice the model memory and per-step compute.

Experiment	Model / width	Task grid	Samples	Steps / LR	Seeds	Primary purpose
E1	dense linear / 96	$\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$	256	500 / .01	4–11	causal phase grid
E2	dense linear / 32–256	(1, 0), (2, 2), (4, 4), (8, 8)	1024	400 / .01	0–7	held-out closure convergence
E-NL	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	20–27	direct-drift crossover certificate
E3	dense linear / 96	$\rho = 4$, lag 2	256	500 / .01	40–47	eight-method CDC ablation

9 Experimental protocol

9.1 Synthetic task and pairing

Sequences have two input channels. A strong cue of scale ρ and a unit weak cue appear at controlled sequence positions separated by a lag. Positive tasks align the cues with the label; the negative control reverses the strong cue. Data and initialization are paired across both-feature and weak-only conditions. Training is deterministic full-batch cross-entropy with zero weight decay. Except for E-NL, the two conditions can be run sequentially because deterministic full-batch updates preserve the same trajectories; E-NL uses lockstep models so both exact drifts are available at each logged time.

E1 uses gain 0.5, $\beta = 0.25$, horizon $H = 5$, and a single eight-seed negative-control cell $(\rho, \text{lag}) = (4, 4)$. E2 uses gain 0.5 and a held-out empirical closure reference with width 384, seeds 100–115, 16 particles, and 32,768 field samples. E-NL uses gain 0.8, cue noise 0.1, zero background noise, $\beta = 0.5$, and direct autograd drift at every logged point. E3 compares ERM, Spectral Decoupling, an interaction penalty, CDC, loss-gradient projection, unconstrained rescue, Bloop-style, and PCGrad-style methods. The five rescue methods share the same weak-only shadow target and differ in what they protect.

Statistics and censoring. Seeds are the unit of replication. Reported confidence intervals are the repository’s 95% seed-level intervals. Hitting times beyond the horizon are right-censored and never converted into finite delays. Conditional crossing-time means are reported only with their number of crossing seeds. The exploratory E-NL 5×5 grid uses seeds 0–3 and is explicitly post-hoc; it is not used to select a confirmatory condition.

Reproducibility. Each run writes a resolved YAML, environment record, trajectories, summary, aggregate tables, and vector/raster figures. The final E-NL rerun is tracked with its resolved configuration and SHA-256 fingerprint over `run_experiment.py` and `src/**/*.*py`; its compact outputs and provenance manifest are under `paper/artifacts`. Historical E1, E2, E2-R, exploratory E-NL, and E3 artifacts record dirty Git states but no source fingerprint, so their exact executed source cannot be reconstructed retroactively from their commit hashes.

10 Results

10.1 Controlled dense-linear phase grid

At lags 0, 2, and 4, all 32 condition–seed pairs per lag have positive weak AUC gaps. Averaged across ρ , the lag-wise gaps are 5.76, 8.93, and 6.11, respectively; the effect peaks at lag 2 rather than decreasing monotonically with lag. At every fixed lag, the gap increases monotonically over the measured ρ grid. The negative-control cell is transfer in 8/8 seeds, with mean gap -4.038 and 95% CI $[-4.498, -3.578]$. Every lag-8 cell is weak-only unlearnable, so these 32 rows are not labeled starvation. Final accuracy is 1.0 throughout, and no row meets the target at initialization.

10.2 Finite-width agreement and exact special cases

E2 compares finite networks with a disjoint-seed empirical *closure reference*; it is not an independently derived theory. Overall both-condition trajectory NRMSE is 0.09084, 0.04761, 0.05149, 0.02958 at widths

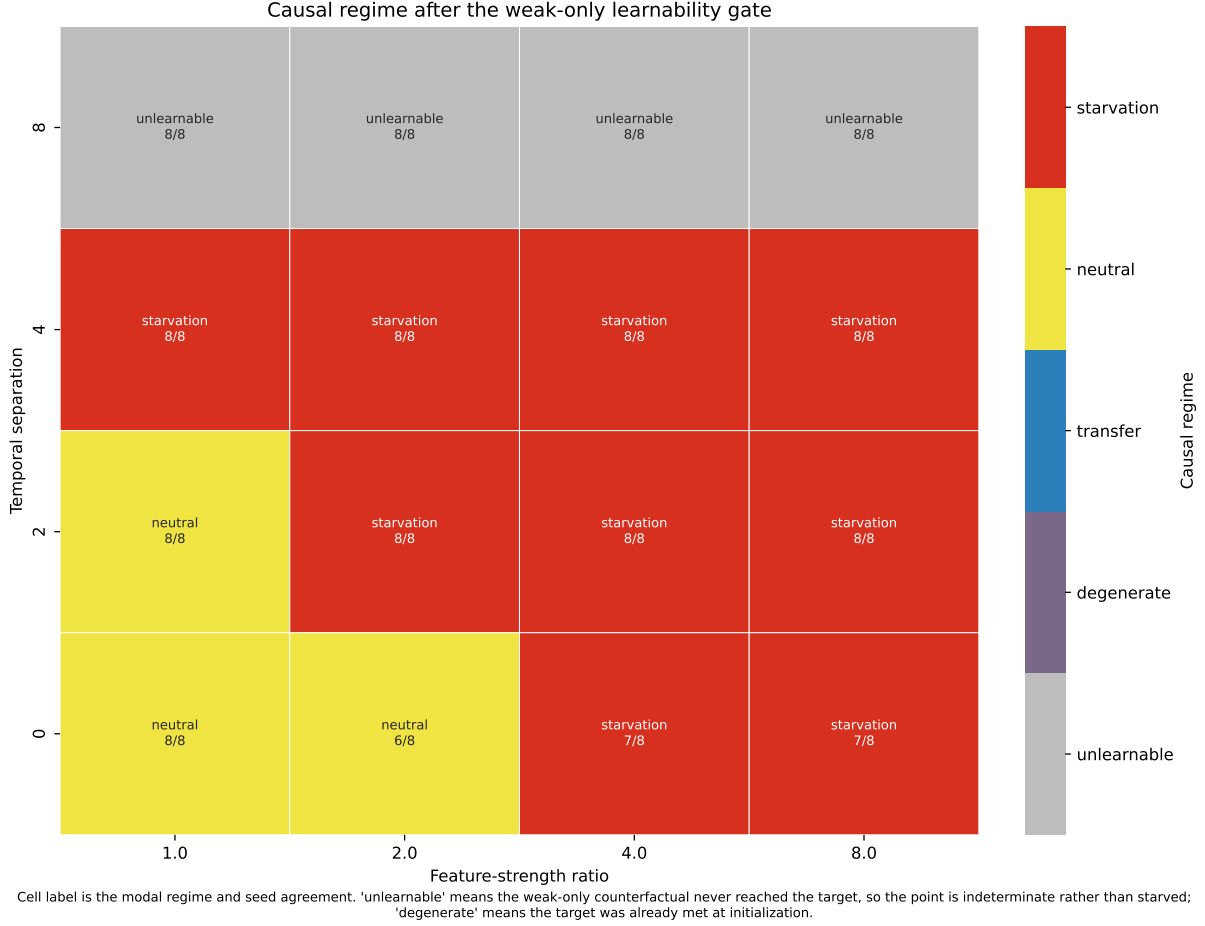


Figure 1: Controlled dense-linear causal regions. The learnability gate prevents lag-8 points from being mislabeled as starvation.

32, 64, 128, 256. Weak-only NRMSE is 1.56931, 0.38792, 0.40201, 0.20024. Both improve end-to-end but are non-monotone at $N = 128$. All 32 point-condition-diagnostic combinations improve end-to-end; large normalized weak-only values at difficult points have near-zero denominators. The exact finite-width geometry check has zero recorded relative error and the maximum projected-identity error is 2.86×10^{-6} .

In the exact zero-disorder solver check, the time-discretization residual scales as $O(\eta)$ with fitted slope 1.0115. A frozen-geometry integrator agrees with the projected-flow integrator to 1.91×10^{-15} . The general solver acceptance record remains false by construction because all general-DMFT checks are blocked.

10.3 Nonlinear crossover is seed-dependent

Table 3 is the authoritative nonlinear result. Tanh has a complete exact-drift/response/tail-area causal certificate in 3/8 seeds; the other five have negative exact relative drift from initialization and therefore do not exhibit transfer-to-starvation. GRU has exact drift and response crossings in 5/8 seeds, but the weak-only trajectory fails to reach $\beta = 0.5$ in all eight seeds; its causal count is therefore zero. The nonzero direct-versus-projected residuals empirically confirm that Gg cannot be silently treated as the exact nonlinear probe derivative.

The post-hoc tanh search over $\rho \in \{1.5, 2, 3, 4, 6\}$ and lags $\{0, 1, 2, 4, 8\}$ found no cell with 4/4 causal certificates; the best cells reached 3/4. We retain this as a negative search audit and do not select a searched cell for a paper claim. In particular, the superseded 8/8 tanh result and its $\tau^* = 1.6112$ used a different response convention and are not valid under the final definition.

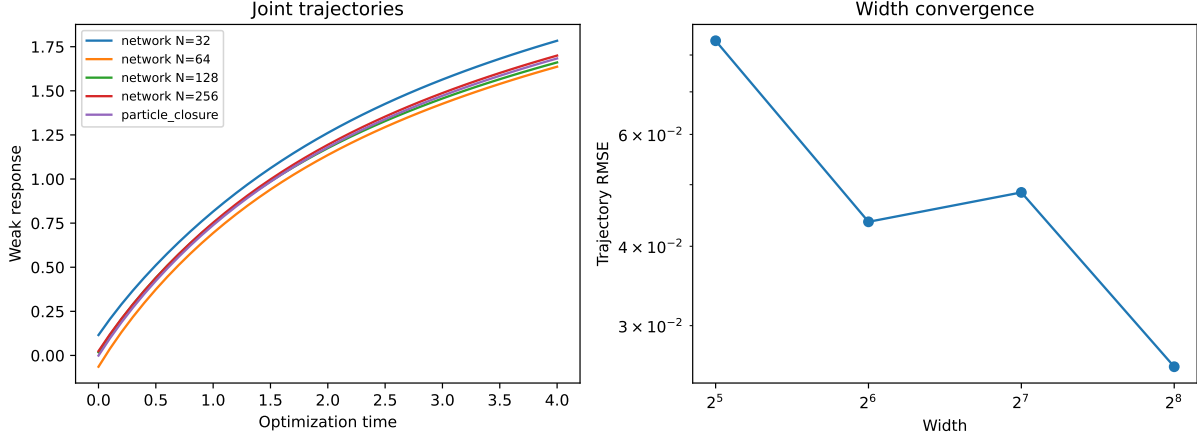


Figure 2: Finite-width agreement with a held-out empirical finite-network closure reference. This is not DMFT validation.

Table 3: Final E-NL result at $\rho = 4$, lag 2, eight seeds. Crossing times and confidence intervals are conditional on crossing and must not be read as eight-seed estimates.

Model	exact drift	response	causal	mean τ_d	mean τ_r	max $ R_w^B - R_w^W $
tanh	3/8	3/8	3/8	0.194	0.389	0.02197
GRU	5/8	5/8	0/8	3.846	7.200	0.002986

10.4 CDC has the local guarantee, not the best final tradeoff

All five shadow-based methods reduce the causal weak AUC gap from ERM’s 10.5155 to approximately zero and attain their shared weak-drift target at every logged step. Their weak outcomes are numerically similar to roughly four decimal places, but not statistically equivalent: 9/10 direct paired comparisons are significant at 0.05, while the differences are practically tiny relative to the roughly 10.5 change from ERM. CDC alone preserves instantaneous strong drift to rounding scale, 4.77×10^{-7} , versus 1.48×10^{-1} to 3.75×10^{-1} for the alternative shadow constraints. This local property does not yield the best final strong response: unconstrained rescue and the PCGrad-style arm end at 1.295, compared with CDC’s 1.042 and ERM’s 1.364.

11 Discussion and limitations

What is proved. The finite-width cross-kernel identity, exact-realization corollary, dense-linear geometry, tail-area equivalence, conditional rank-one crossover criterion, transverse hitting-time bound, singular zero-disorder wide limit, and local CDC results are mathematical statements with proofs. Executable contracts reproduce their algebra and reject scope violations.

What the experiments establish. The controlled dense model shows robust positive AUC gaps in the learnable region. The nonlinear experiment does not establish a universal crossover: tanh is seed-dependent, and GRU fails causal learnability despite response crossings. The exploratory search did not find a 4/4 robust cell. CDC demonstrates its exact local constraint but not a superior practical tradeoff.

Open mean-field theory. For positive recurrent disorder or positive lag, a joint optimization-time DMFT would need (i) an effective single-site process and covariance/response kernels, (ii) all visible and loss-invisible closure variables, (iii) finite-horizon existence and uniqueness, (iv) joint quenched concentration under shared

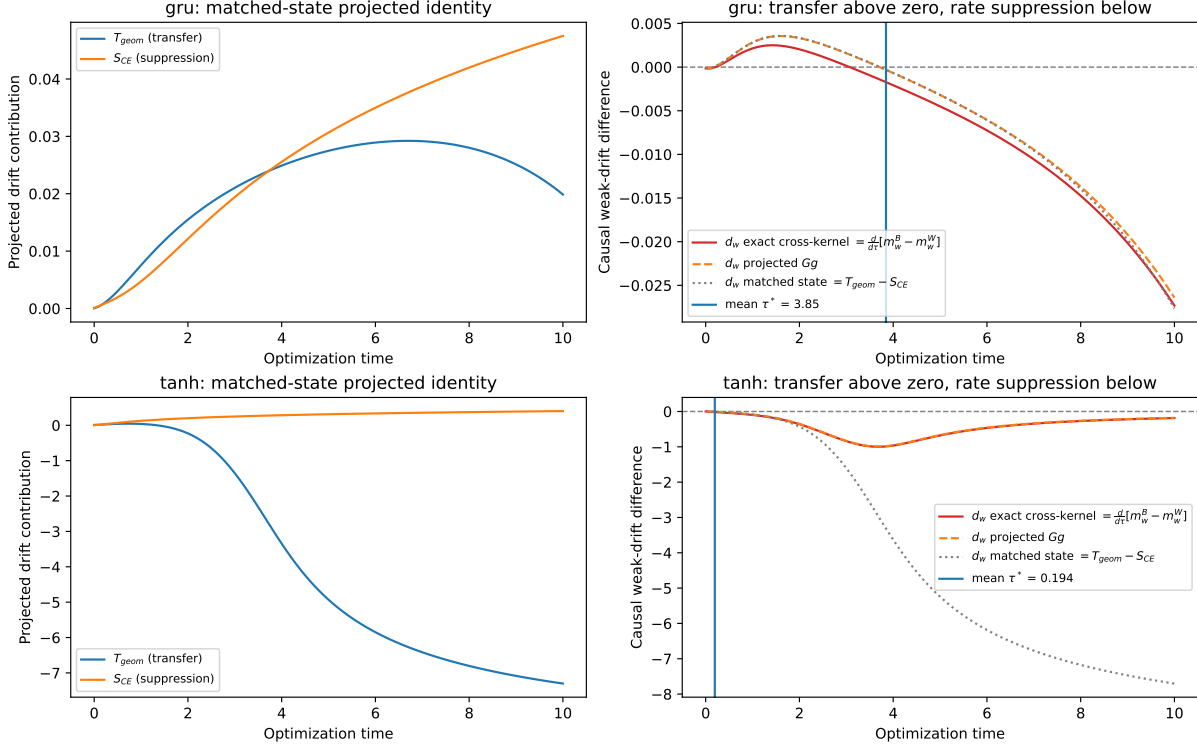


Figure 3: Theorem-aligned nonlinear trajectories. Direct-autograd drift is primary; projected and matched-state quantities are separately styled diagnostics.

disorder, and (v) discretization and hitting-time control. None is supplied by fitting an empirical closure reference. We therefore leave an analytic ρ_c and independent tanh/GRU τ^* prediction open.

Data and external validity. All citable results are controlled synthetic experiments. A Waterbirds adapter exists but has not been run. Its proposed weak coordinate is an intercept rather than an identified bird-shape response, and no exact background ablation exists; we make no Waterbirds claim. If pursued, the preregistered sensitivity pilot must precede a full run and oracle/group-agnostic information settings must be separated.

Reproducibility and baselines. The final E-NL rerun is content-addressed and compactly archived, but historical E1, E2, E2-R, exploratory E-NL, and E3 runs retain a disclosed dirty-source provenance limitation. The Bloop-style and PCGrad-style baselines are repository implementations motivated by the cited methods, not yet independently audited replicas. CDC also has a stronger supervision and compute setting than ERM, Spectral Decoupling, and the interaction penalty.

Submission readiness. The package is scientifically defensible as a controlled theory paper, but the seed-dependent nonlinear headline, lack of real-data evidence, historical dirty-run provenance, and unaudited external baselines limit its present empirical strength. No theorem or experiment can guarantee an oral acceptance outcome.

12 Conclusion

Gradient starvation requires more than a decreasing weak-feature rate. In a paired causal comparison, a rate crossover becomes an outcome crossing exactly when its negative tail area repays the earlier transfer

Table 4: Dense CDC ablation, eight paired seeds. Shadow-based arms have access to a matched weak-only model; the other arms do not. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	Shadow	Weak AUC gap	Strong change	Final m_s
ERM	no	10.5155	—	1.364
Spectral Decoupling	no	9.0229	—	0.829
Interaction	no	10.5145	—	1.365
CDC	yes	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	yes	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	yes	−0.0007	1.48×10^{-1}	1.295
Bloop-style	yes	0.0002	3.75×10^{-1}	0.370
PCGrad-style	yes	−0.0007	1.48×10^{-1}	1.295

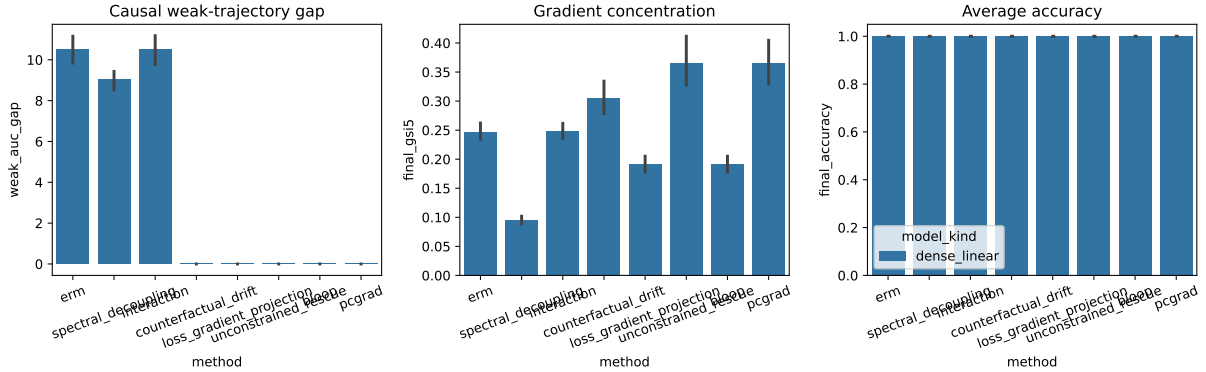


Figure 4: Eight-method dense-linear mitigation ablation. CDC’s established distinction is instantaneous first-order preservation, not superior weak rescue or final strong retention.

advantage, and a causal label also requires weak-only learnability. This distinction turns an ambiguous mechanism into a theorem and an executable certificate. The same discipline narrows the rest of the paper: $\dot{M} = Gg$ is exact only under explicit realization assumptions; the rank-one criterion predicts only when its bounds are independently established; the wide limit is singular; and CDC’s exact guarantee is local. The corrected nonlinear results are correspondingly modest. Establishing a general recurrent optimization-time theory remains an open problem rather than a numerical placeholder.

A Proofs for the finite-width theorem package

A.1 Proof of Theorem 1 and Corollary 2

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}_C = -\nabla L_C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (34)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}_C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (35)$$

which proves Theorem 1. Under exact realization, $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C$. Substitution and exchange of finite sums yield Eq. (7). If an orthogonal residual logit term is present, substitution of $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C + \nabla r_i^\perp$ instead yields Eq. (8). $\square$

A.2 Proof of the dense-linear geometry

For a unit response mode in channel a at lag ℓ_a , write b_a for the corresponding input column. Direct differentiation of $c^\top W^{\ell_a} b_a$ gives gradient $p_a = W^{\ell_a} b_a$ with respect to c , gradient $q_a = (W^{\ell_a})^\top c$ with respect to column b_a , and recurrent gradient

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (36)$$

The parameter blocks are disjoint, so their gradient inner products add. Input columns contribute only when the channels coincide, giving Eq. (9). $\square$

A.3 Proof of Theorem 3 and Corollary 4

Absolute continuity and the fundamental theorem of calculus give

$$\Delta(\tau_d) = \Delta(0) + \int_0^{\tau_d} d = A_+, \quad \Delta(t) = \Delta(\tau_d) + \int_{\tau_d}^t d = A_+ - N(t). \quad (37)$$

The sign assumption makes N continuous and strictly increasing on every nontrivial post-crossing interval. Equality, strict negativity, and uniqueness are therefore respectively equivalent to $N(H) \geq A_+$, $N(H) > A_+$, and strict post-crossing growth of N .

If $d \leq -\kappa$ after t_0 , then $N(t) - N(t_0) \geq \kappa(t - t_0)$; solve for the time needed to cover the remaining area. If $d'(t) \leq -\lambda$ and $d(\tau_d) = 0$, integration gives $d(t) \leq -\lambda(t - \tau_d)$ and then $N(t) \geq \lambda(t - \tau_d)^2/2$. Applying the theorem proves both stated bounds. $\square$

A.4 Proof of Theorem 5 and Corollary 6

The deterministic cue law gives $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$. Equation (7) therefore gives $F_w^B = s_B H_B$, $F_w^W = s_W H_W$, $u'_B = s_B Q_B$, and $u'_W = s_W H_W$. Since

$$\frac{d}{du} \log \sigma(-u) = -(1 - \sigma(-u)), \quad (38)$$

differentiating Ψ yields Eq. (19). When both drifts are positive, monotonicity of the logarithm gives $\text{sign}(F_w^B - F_w^W) = \text{sign}(\Psi)$.

Under $\Psi' \leq -\kappa$, integration gives $\Psi(t) \leq \Psi(t_0) - \kappa(t - t_0)$. Continuity forces a zero by $t_0 + \Psi(t_0)/\kappa \leq H$, while strict decrease gives uniqueness. The primitive inequalities imply the derivative bound term by term. $\square$

A.5 Proof of Corollary 7

Transversality implies locally before τ that $x(t) \leq \beta - \kappa(\tau - t)$ and after τ that $x(t) \geq \beta + \kappa(t - \tau)$. Uniform error excludes an x_N crossing before $\tau - \epsilon_N/\kappa$ and forces one by $\tau + \epsilon_N/\kappa$. Separation excludes a spurious earlier crossing. A sampling grid adds at most one spacing, and the paired bound follows by the triangle inequality. $\square$

A.6 Proof of Theorems 8 and 9

At zero recurrent gain and lag, gradient flow gives $\dot{b}_a = g_a c$ and $\dot{c} = \sum_b g_b b_b$. Differentiating $M_a = c \cdot b_a$, $u_{ab} = b_a \cdot b_b$, and $w = c \cdot c$ proves Eq. (25). The mode gradient has one component in the c block and one in its own b_a block, yielding $G_{ab} = u_{ab} + \delta_{ab} w$.

At initialization, the variances of M_s, M_w, u_{sw} are $1/N$ and those of $u_{ss} - 1, u_{ww} - 1, w - 1$ are $2/N$. Coordinatewise Chebyshev bounds and a union bound sum to $1 + 1 + 1 + 2 + 2 + 2 = 9$, proving Eq. (27).

For bounded second cue moments the CE field is locally Lipschitz. Define the Gram energy $R = w + \text{tr}(u)$. Along the six-scalar flow,

$$R' = 4g \cdot M = 4\mathbb{E}[X\sigma(-X)], \quad X = z \cdot M. \quad (39)$$

The scalar $x\sigma(-x)$ is bounded above and $R \geq 0$, so R cannot blow up on a finite horizon. Cauchy–Schwarz bounds all coordinates by R , yielding a compact forward set and a finite local Lipschitz constant on a common tube. Gronwall’s inequality gives Eq. (28). Substitute $\epsilon = \delta e^{-L_H H}$ into Eq. (27) to obtain Eq. (29). The shared-initialization paired result uses the larger of the two finite-horizon Lipschitz constants. $\square$

B Proofs for Counterfactual Drift Correction

B.1 Instantaneous preservation

If $a = 0$, both strong drifts vanish. Otherwise,

$$a \cdot q = a \cdot p - \frac{p \cdot a}{\|a\|^2} (a \cdot a) = a \cdot p - p \cdot a = 0, \quad (40)$$

so $a \cdot v' = a \cdot v + \alpha a \cdot q = a \cdot v$. This argument is independent of α , hence survives a cap. $\square$

B.2 Unique minimum-norm correction

If $\delta \leq 0$, $d = 0$ is feasible and uniquely norm-minimal. Suppose $\delta > 0$. Any feasible d obeys $a \cdot d = 0$ and $p \cdot d \geq \delta$. Because $p = q + (p \cdot a / \|a\|^2) a$ when $a \neq 0$ (and $q = p$ when $a = 0$), $p \cdot d = q \cdot d$. Cauchy–Schwarz on the subspace orthogonal to a gives

$$\delta \leq q \cdot d \leq \|q\| \|d\|, \quad \|d\| \geq \delta / \|q\|. \quad (41)$$

Equality requires a nonnegative multiple of q ; the unique target-attaining choice is $d^* = (\delta / \|q\|^2) q$. A positive deficit is feasible exactly when $q \neq 0$. $\square$

B.3 Local finite-step deviation

For an L -Lipschitz gradient on both segments, the smoothness remainder satisfies

$$|m_s(\theta + h) - m_s(\theta) - \nabla m_s(\theta) \cdot h| \leq \frac{L}{2} \|h\|^2. \quad (42)$$

Apply this at $h = \eta v$ and $h = \eta v'$. Instantaneous preservation cancels the linear terms and the triangle inequality yields Eq. (32). For active uncapped CDC,

$$\|v'\| \leq \|v\| + \frac{\delta}{\|q\|} \leq V + D/q_{\min}, \quad (43)$$

which gives the explicit bound. If smoothness is known only on $B(\theta, r)$, sufficient segment conditions are $\eta V \leq r$ and $\eta(V + D/q_{\min}) \leq r$. $\square$

C Discrete certificate and experiment audit

For logged values Δ_k , increments telescope exactly:

$$\Delta_K - \Delta_0 = \sum_{k=0}^{K-1} (\Delta_{k+1} - \Delta_k). \quad (44)$$

The executable certificate requires one strict block of positive increments followed by one strict block of negative increments. The peak-minus-initial value is its positive area; peak-minus-final is its negative tail area. It certifies causal starvation only if the final trajectory is strictly below the initial gap and the external weak-only learnability flag is true. This finite-step certificate makes no assertion between logged points.

Table 5: Authoritative artifact map. Paths are relative to the repository root.

Claim family	Artifact directory
Controlled dense phase grid	<code>results/e1.dense_rerun-20260823-085142</code>
Finite-width empirical closure	<code>results/e2.corrected_validation-20260821-224630</code>
Exact special solver checks	<code>results/e2r_solver_checks-20260823-084911</code>
Final nonlinear crossover	<code>paper/artifacts/enl_tanh_crossover-20260824-141207</code>
Exploratory nonlinear audit	<code>results/enl_exact_regime_search-20260824-105003</code>
Dense CDC ablation	<code>results/e3.cdc.dense.ablation-20260823-090702</code>

Reproduction commands. From the repository root in the project environment:

```
python run_experiment.py e1 --config configs/e1_dense_rerun.yaml
python run_experiment.py e2 --config configs/e2_corrected_validation.yaml
python run_experiment.py e2r --config configs/e2r_solver_checks.yaml
python run_experiment.py enl --config configs/enl_tanh_crossover.yaml
python run_experiment.py e3 --config configs/e3_cdc_dense_ablation.yaml
```

The exploratory E-NL command is deliberately omitted from the confirmatory list; its complete configuration and negative result remain archived in the repository.

References

- David G. Clark, Blake Bordelon, Jacob A. Zavatone-Veth, and Cengiz Pehlevan. Structure, disorder, and dynamics in task-trained recurrent neural circuits. *bioRxiv*, 2026. doi: 10.64898/2026.03.02.708943. URL <https://www.biorxiv.org/content/10.64898/2026.03.02.708943v1>.
- Yoav Ger and Omri Barak. Learning reveals invisible structure in low-rank RNNs. *arXiv preprint arXiv:2605.04115*, 2026. URL <https://arxiv.org/abs/2605.04115>.
- Yu-Guan Hsieh, James Thornton, Eugene Ndiaye, Michal Klein, Marco Cuturi, and Pierre Ablin. Careful with that scalpel: Improving gradient surgery with an EMA. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 19085–19100, 2024. URL <https://proceedings.mlr.press/v235/hsieh24a.html>.
- Mohammad Pezeshki, Sékou-Oumar Kaba, Yoshua Bengio, Aaron Courville, Doina Precup, and Guillaume Lajoie. Gradient starvation: A learning proclivity in neural networks. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL <https://openreview.net/forum?id=aExAsh1UHZo>.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/3fe78a8acf5fda99de95303940a2420c-Abstract.html>.